

How to Handle Sequential Data?

Definition of Sequential Data

Sequential data refers to any dataset where the order of the elements matters, meaning that each data point depends on previous ones. Unlike independent data points (e.g., images in computer vision), sequential data contains a natural progression.

Examples of Sequential Data

- **Time-Series Data:** Stock prices, weather forecasts, sensor readings.
- **Natural Language Processing (NLP):** Sentences, speech, chatbots.
- **Video Processing:** Frame-by-frame analysis (e.g., action recognition).
- **Music & Audio:** Speech recognition, melody prediction.
- **Biological Sequences:** DNA/RNA sequences in bioinformatics.

Definition of a Sequential Task

A **sequential task** is a machine learning problem that requires modeling dependencies across time (or ordered steps). The key characteristic is that earlier inputs influence later outputs.

Examples of Sequential Tasks

- **Time-Series Forecasting:**
 - Predicting the next day's stock price based on past prices.
 - Weather forecasting using past temperature and humidity data.
- **Speech Recognition:**

- Converting spoken words into text (e.g., Siri, Alexa).
- The system needs to remember past words to understand the sentence correctly.
- **Machine Translation:**
 - Translating “Je t’aime” → “I love you” requires understanding sentence structure.
- **Chatbots & Language Models:**
 - In chatbots (e.g., GPT models), responses depend on past conversation history.
- **Action Recognition in Videos:**
 - Identifying whether a person is running or jumping based on previous frames.

Why Do Sequential Tasks Need Special Models?

- In traditional **feedforward neural networks (FFNs)**, all inputs are processed independently, meaning they lose order information.
- **Recurrent Neural Networks (RNNs), LSTMs, GRUs, and Transformers** were designed to handle sequences by remembering previous states while processing new inputs.

Key Takeaways

- **Sequential data has an inherent order** that affects the next step.
- **Sequential tasks require memory** to retain past information.
- **Standard FFNs fail** because they treat inputs as independent.
- **RNNs, LSTMs, and Transformers** solve this by incorporating memory mechanisms.